

Redwan Ahmed Rizvee

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

PhD in Computer Science (Advisor: Dr. Alexandra Meliou)

Sep. 2025 – Present

- Focus: Data Mining, Perception-Aware Sampling, Visual Analytics, Database Theory.

University of Dhaka

Dhaka, Bangladesh

Master of Science in Computer Science and Engineering

Jan. 2019 – Sep. 2021

- CGPA: **4.00/4.00** (Ranked 1st in class)

University of Dhaka

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

Jan. 2015 – Dec. 2018

- CGPA: **3.85/4.00** (Ranked 1st in class) | *Dean's Award, Gold Medalist*

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, JavaScript (ES6+), Shell Scripting

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy

Web & Frameworks: Django, Flask, React, Node.js, HTML5/CSS3, Bootstrap

Tools & Platforms: Git, Docker, Linux/Unix, MySQL, PostgreSQL, Firebase, LaTeX

EXPERIENCE

University of Massachusetts Amherst (DREAM Lab)

Amherst, MA

Graduate Research Assistant

Sep. 2025 – Present

- Developing **Perception-Aware Sampling** algorithms to optimize visual fidelity in large-scale scatter plots.
- Designing statistical methods to preserve outliers and dense regions, ensuring high data coverage and representational accuracy while addressing computational efficiency in big data visualization.

Cognitive Agents and Interaction Lab (CAIL)

Dhaka, Bangladesh

Co-Investigator

Jan. 2023 – Aug. 2025

- Supervised projects utilizing Deep Learning and Reinforcement Learning for agent-based decision making.
- Architected **COLAS** (Coding Lab Assistant), an automated source code evaluation system that integrates plagiarism detection and strict monitoring for academic labs.

TigerIT Bangladesh Limited

Dhaka, Bangladesh

Research Scientist

July 2020 – Feb. 2022

- Led R&D on statistical sensor data analysis and deployed rule-based algorithms for production environments.
- Designed and implemented **Encoder-Decoder architectures with Attention** mechanisms for audio processing and Natural Language Processing (NLP) tasks.
- Optimized deep learning pipelines, resulting in improved inference speed and model accuracy for different applications.

Data Mining Research Lab

Dhaka, Bangladesh

Research Assistant

Jan. 2018 – Jul 2025

- Developed novel pattern mining algorithms focusing on scalability and resource efficiency for time-series data.
- Formulated metrics for robust correlation analysis and cluster generation in high-dimensional datasets.

Data and Design Lab (DnD Lab)

Dhaka, Bangladesh

Research Assistant

Apr. 2019 – Sep. 2019

- Worked in the BDQOL project, which focuses on developing a composite indicator to measure the quality of life index for low-resource countries. I contributed to the research, data collection, data processing, and visual dashboard implementation.

SELECTED PROJECTS

- COLAS - Coding Lab Assistant** | *Python, React, Django, LLM* [COLAS]
- Built a scalable, full-stack automated evaluation system for live programming labs.
 - Integrated plagiarism detection and real-time monitoring, improving grading efficiency for large student cohorts.
- Bangladesh Quality of Life Indicator (BDQOL)** | *JavaScript, Data Viz* [BDQOL]
- Developed an interactive visualization tool to track socio-economic conditions based on SDG indicators.
 - Handled data pipeline engineering to process complex socio-economic datasets for web rendering.
- MSCLM-A** | *Python, NLP*
- Created a memory-flexible, language-independent sentence completion module using Arpa-based Language Models.

SELECTED PUBLICATIONS

(Full list of 15+ publications available on Google Scholar)

- R. A. Rizvee**, C. F. Ahmed, M. F. Arefin, C. K. Leung, “A new tree-based approach to mine sequential patterns”, *Expert Systems with Applications*, vol. 242, 122754, 2024. [DOI]
- R. A. Rizvee**, R. Hassan, M. M. Khan, “Reinforcement Learning-Based Formulations With Hamiltonian-Inspired Loss Functions for Combinatorial Optimization”, *IEEE Access*, 2024.
- R. A. Rizvee**, et al., “LeafNet: A proficient convolutional neural network for detecting seven prominent mango leaf diseases”, *Journal of Agriculture and Food Research*, 2023.

HONORS & AWARDS

- Research Excellence Recognition (2025):** University of Dhaka, for exemplary scholarly contribution.
- Dean’s Award & Gold Medal (2019):** For achieving the highest distinction in the Faculty of Engineering.
- Fellowship Scholarship:** ICT Division, Bangladesh, for high-impact research.